

Salifort Motors | Executive Summary

Employee Turnover Prediction and Retention Insights

Business problem

The case study analyzes employee turnover patterns and builds a model to identify employees who may be at higher risk of leaving, supporting HR investigation and retention planning.

Data

14,999 employee records and 10 variables covering satisfaction, performance evaluation, project load, monthly working hours, tenure, work accidents, promotion history, department, salary, and turnover. The dataset is educational and does not represent a real Salifort Motors workforce.

Analytical approach

- Checked structure, missing values, duplicates, and distributions.
- Removed 3,008 duplicate records before the main analysis.
- Explored workload, projects, tenure, evaluation, salary, and department patterns.
- Built a logistic regression baseline and compared tree-based approaches.
- Engineered an overworked indicator using more than 175 average monthly hours.
- Evaluated the final feature-engineered Random Forest on the saved holdout set.
- Interpreted feature importance and translated signals into HR investigation areas.

Final holdout result

Metric	Random Forest
Accuracy	96.2%
Precision	87.0%
Recall	90.4%
F1-score	88.7%
AUC	93.8%

Key signals

The saved feature-importance analysis highlights last_evaluation, number_project, tenure, and overworked among the strongest predictive variables. These are predictive signals within this dataset, not causal explanations.

Practical implications

- Review workload and project allocation for employees carrying many projects or long working hours.
- Examine career-development and promotion patterns alongside tenure.
- Investigate employees with high evaluation scores who also work very long hours.
- Use model outputs to prioritize human review and follow-up, not automated employment decisions.

Limitations and next step

The dataset is educational, and the notebook reused a holdout workflow across iterative modeling rounds. Therefore the reported holdout metrics should not be treated as an independent production performance estimate. Validate the strongest patterns on fresh workforce data before operational use.